Nonlinear Modelling, Identification and Control of UUVs

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2.1 Introduction

This chapter presents the UUV equations of motion using the results of Fossen [1][2]. The nonlinear model presented in this chapter is mainly intended for control systems design in combination with system identification and parameter estimation. The resulting model is decoupled into longitudinal and lateral motions such that autopilots for speed, depth/diving, and heading control can be designed.

The motivation for using non-linear theory is that one model can cover the whole flight envelope of the UUV, instead of linearising the model about many working points and using gain scheduling between these. In addition, physical model properties, in terms of first principles are not destroyed which is a primary problem associated with linearising a plant. Finally, non-linear optimisation methods are used as a tool for system identification, while PID-control and backstepping demonstrate how non-linear autopilots can be designed.

2.1.1 Notation

In the forthcoming chapter, vectors of dimension n in the *Euclidean space* are denoted as $\mathbf{x} \in \mathbb{R}^n$, matrices are denoted as $\mathbf{A} \in \mathbb{R}^{m \times n}$ and scalars are written as $a \in \mathbb{R}$.

A matrix $\mathbf{S} \in SS(n)$, that is the set of skew-symmetric matrices of order n, is said to be skew-symmetrical if:

$$\mathbf{S} = -\mathbf{S}^\top$$

This implies that the off-diagonal elements of $\mathbf{S}$ satisfy $s_{ij} = -s_{ji}$ for $i \neq j$ while the diagonal elements are zero.

The vector cross product $\times$ is defined by:

$$\lambda \times \mathbf{a} := \mathbf{S}(\lambda)\mathbf{a} \tag{2.1}$$

where $\mathbf{S} \in SS(3)$ is defined as:

$$\mathbf{S}(\lambda) = -\mathbf{S}^\top(\lambda) = \begin{bmatrix} 0 & -\lambda_3 & \lambda_2 \\ \lambda_3 & 0 & -\lambda_1 \\ -\lambda_2 & \lambda_1 & 0 \end{bmatrix}, \quad \lambda = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} \tag{2.2}$$

For marine vehicles moving in 6 *degrees of freedom* (DOF), 6 independent coordinates are necessary to determine the position and orientation. The first three coordinates, and their time derivatives, correspond to the position and translational motion along the x-, y-, and z-axes respectively, while the last 3 coordinates, and their time derivatives, are used to describe orientation and rotational motions. The 6 motion components are conveniently defined as *surge, sway, heave, roll, pitch* and *yaw* (SNAME [3]).

The general motion of an UUV in 6 DOF is modelled by using the notation of Fossen [4]. Consider the following vectors:

$$\eta = [n, e, d, \phi, \theta, \psi]^\top, \quad \nu = [u, v, w, p, q, r]^\top \tag{2.3}$$

where n, e, d denote the North-East-Down positions in Earth-fixed coordinates (n-frame), ϕ, θ, ψ are the $C_{D\beta\beta}$ angles (orientation), and u, v, w, p, q, r are the 6 DOF generalised velocities in body-fixed coordinates (b-frame).

2.2 Modelling of UUVs

The UUV equations of motion involve the study of two fields of mechanics: *statics* and *dynamics*. Statics deals with the forces that produce equilibrium, such as buoyancy or gravity, while dynamics focuses on the analysis of forces primarily in the context of the motions produced in bodies.

Statics is the oldest of the engineering sciences, dating back to Archimedes (287–212 BC), who derived the basic law of hydrostatic buoyancy. The study of dynamics started much later, since accurate measurements of time are necessary to perform dynamic experiments. The scientific basis of dynamics was provided by Sir Isaac Newton's Principia, published in 1687.

The study of *dynamics* can be divided into two parts: *kinematics* (Section 2.2.1), which treats only geometric aspects of motion, without consideration of mass or forces, while *kinetics* (Section 2.2.2) is the analysis of the forces causing these motions.

2.2.1 6 DOF Kinematic Equations

In Fossen [1][2] it was shown that the 6 DOF kinematic equations can be expressed in vector form as:

$$\dot{\eta} = \mathbf{J}(\eta)\nu \tag{2.4}$$

where

$$\mathbf{J}(\eta) = \begin{bmatrix} \mathbf{R}_b^n(\mathbf{\Theta}) & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{T}_\Theta(\mathbf{\Theta}) \end{bmatrix} \tag{2.5}$$

with $\eta \in \mathbb{R}^3 \times \mathcal{S}^3$ (torus of dimension 3 implying that there are three angles defined on the interval $[0, 2\pi]$) and $\nu \in \mathbb{R}^6$. The Euler angle rotation matrix $\mathbf{R}_b^n(\mathbf{\Theta}) \in \mathbb{R}^{3\times3}$ is defined in terms of the

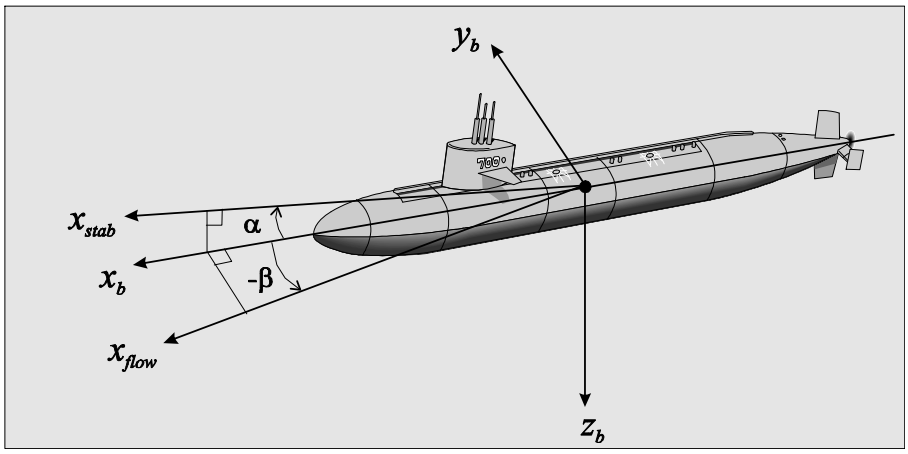

Figure 2.1: Submarine.

principal rotations:

$$\mathbf{R}_{x,\phi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \mathrm{c}\phi & -\mathrm{s}\phi \\ 0 & \mathrm{s}\phi & \mathrm{c}\phi \end{bmatrix}, \quad \mathbf{R}_{y,\theta} = \begin{bmatrix} \mathrm{c}\theta & 0 & \mathrm{s}\theta \\ 0 & 1 & 0 \\ -\mathrm{s}\theta & 0 & \mathrm{c}\theta \end{bmatrix}, \quad \mathbf{R}_{z,\psi} = \begin{bmatrix} \mathrm{c}\psi & -\mathrm{s}\psi & 0 \\ \mathrm{s}\psi & \mathrm{c}\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2.6}$$

where $\mathrm{s}\,\cdot = \sin(\cdot)$ and $\mathrm{c}\,\cdot = \cos(\cdot)$ using the zyx-convention:

$$\mathbf{R}_b^n(\mathbf{\Theta}) := \mathbf{R}_{z,\psi}\mathbf{R}_{y,\theta}\mathbf{R}_{x,\phi} \tag{2.7}$$

or

$$\mathbf{R}_b^n(\mathbf{\Theta}) = \begin{bmatrix} \mathrm{c}\psi\mathrm{c}\theta & -\mathrm{s}\psi\mathrm{c}\phi + \mathrm{c}\psi\mathrm{s}\theta\mathrm{s}\phi & \mathrm{s}\psi\mathrm{s}\phi + \mathrm{c}\psi\mathrm{c}\phi\mathrm{s}\theta \\ \mathrm{s}\psi\mathrm{c}\theta & \mathrm{c}\psi\mathrm{c}\phi + \mathrm{s}\phi\mathrm{s}\theta\mathrm{s}\psi & -\mathrm{c}\psi\mathrm{s}\phi + \mathrm{s}\theta\mathrm{s}\psi\mathrm{c}\phi \\ -\mathrm{s}\theta & \mathrm{c}\theta\mathrm{s}\phi & \mathrm{c}\theta\mathrm{c}\phi \end{bmatrix} \tag{2.8}$$

The inverse transformation satisfies:

$$\mathbf{R}_b^n(\mathbf{\Theta})^{-1} = \mathbf{R}_n^b(\mathbf{\Theta}) = \mathbf{R}_{x,\phi}^\top\mathbf{R}_{y,\theta}^\top\mathbf{R}_{z,\psi}^\top \tag{2.9}$$

The Euler angle attitude transformation matrix is:

$$\mathbf{T}_\Theta(\mathbf{\Theta}) = \begin{bmatrix} 1 & \mathrm{s}\phi\mathrm{t}\theta & \mathrm{c}\phi\mathrm{t}\theta \\ 0 & \mathrm{c}\phi & -\mathrm{s}\phi \\ 0 & \mathrm{s}\phi/\mathrm{c}\theta & \mathrm{c}\phi/\mathrm{c}\theta \end{bmatrix} \implies \mathbf{T}_\Theta^{-1}(\mathbf{\Theta}) = \begin{bmatrix} 1 & 0 & -\mathrm{s}\theta \\ 0 & \mathrm{c}\phi & \mathrm{c}\theta\mathrm{s}\phi \\ 0 & -\mathrm{s}\phi & \mathrm{c}\theta\mathrm{c}\phi \end{bmatrix}, \quad \theta \neq \pm 90^o \tag{2.10}$$

Notice that $\mathbf{T}_\Theta(\mathbf{\Theta})$ is undefined for a pitch angle of $\theta = \pm 90^o$ and that $\mathbf{T}_\Theta^{-1}(\mathbf{\Theta}) \neq \mathbf{T}_\Theta^\top(\mathbf{\Theta})$. For UUVs operating close to this singularity two Euler angle representations with different singularities can be used to avoid the singular point by simply switching between these. Another possibility is to use a quaternion representation (Fossen [1][2]), which approaches this problem using a four parameter method.

The transformation matrix $\mathbf{J}$ and its elements $\mathbf{J}_1 = \mathbf{R}_b^n(\mathbf{\Theta})$ and $\mathbf{J}_2 = \mathbf{T}_\Theta(\mathbf{\Theta})$ can be computed in Matlab using the MSS toolbox [5]:

$$[\mathtt{J}, \mathtt{J1}, \mathtt{J2}] \;=\; \mathtt{eulerang}(\mathtt{phi}, \mathtt{theta}, \mathtt{psi}).$$

Alternatively, we can write the equations (2.4) in component form as:

$$\begin{aligned}
\dot{n} &= u\cos\psi\cos\theta + v(\cos\psi\sin\theta\sin\phi - \sin\psi\cos\phi) \\
&\quad + w(\sin\psi\sin\phi + \cos\psi\cos\phi\sin\theta) && (2.11)\\
\dot{e} &= u\sin\psi\cos\theta + v(\cos\psi\cos\phi + \sin\phi\sin\theta\sin\psi) \\
&\quad + w(\sin\theta\sin\psi\cos\phi - \cos\psi\sin\phi) && (2.12)\\
\dot{d} &= -u\sin\theta + v\cos\theta\sin\phi + w\cos\theta\cos\phi && (2.13)\\
\dot{\phi} &= p + q\sin\phi\tan\theta + r\cos\phi\tan\theta && (2.14)\\
\dot{\theta} &= q\cos\phi - r\sin\phi && (2.15)\\
\dot{\psi} &= q\frac{\sin\phi}{\cos\theta} + r\frac{\cos\phi}{\cos\theta}, \quad \theta \neq \pm 90^o && (2.16)
\end{aligned}$$

2.2.2 Kinetics

For UUVs it is desirable to derive the equations of motion for an arbitrary origin in the b-frame to take advantage of the vessel's geometric properties. Since the hydrodynamic forces and moments are given in the b-frame, Newton's laws will be formulated in the b-frame as well. The b-frame coordinate system rotates with respect to the n-frame, and the velocities in the b- and n-frame are related through (2.4).

2.2.3 Equations of Motion

In Fossen [1], [2] it is shown that Newton's 2nd law can be expressed in an arbitrary body-fixed coordinate frame as:

$$\mathbf{M}_{RB}\dot{\nu} + \mathbf{C}_{RB}(\nu)\nu = \tau_H + \tau \tag{2.17}$$

where $\tau_H = [X_H, Y_H, Z_H, K_K, M_H, N_H]^\top$ is a vector of the hydrostatic and hydrodynamic forces and moments and τ is a vector of control inputs. The matrices are:

$$\begin{aligned}
\mathbf{M}_{RB} = \mathbf{M}_{RB}^\top &= \begin{bmatrix} m\mathbf{I}_{3\times3} & -m\mathbf{S}(\mathbf{r}_g^b) \\ m\mathbf{S}(\mathbf{r}_g^b) & \mathbf{I}_o \end{bmatrix} \\
&= \begin{bmatrix}
m & 0 & 0 & 0 & mz_g & -my_g \\
0 & m & 0 & -mz_g & 0 & mx_g \\
0 & 0 & m & my_g & -mx_g & 0 \\
0 & -mz_g & my_g & I_x & -I_{xy} & -I_{xz} \\
mz_g & 0 & -mx_g & -I_{yx} & I_y & -I_{yz} \\
-my_g & mx_g & 0 & -I_{zx} & -I_{zy} & I_z
\end{bmatrix}
\end{aligned} \tag{2.18}$$

$$\mathbf{C}_{RB}(\nu) = \begin{bmatrix} \mathbf{0}_{3\times3} & -m\mathbf{S}(\nu_1) - m\mathbf{S}(\nu_2)\mathbf{S}(\mathbf{r}_g^b) \\ -m\mathbf{S}(\nu_1) + m\mathbf{S}(\mathbf{r}_g^b)\mathbf{S}(\nu_2) & -\mathbf{S}(\mathbf{I}_o\nu_2) \end{bmatrix}, \tag{2.19}$$

where $\nu_1 = [u, v, w]^\top$, $\nu_2 = [p, q, r]^\top$, $\mathbf{r}_g^b = [x_g, y_g, z_g]$ is the centre of gravity (CG) with respect

to the b-frame origin, and:

$$\mathbf{I}_o := \begin{bmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{yx} & I_y & -I_{yz} \\ -I_{zx} & -I_{zy} & I_z \end{bmatrix}, \qquad \mathbf{I}_o = \mathbf{I}_o^\top > 0 \tag{2.20}$$

is the *inertia matrix* about the b-frame origin. The rigid-body dynamics can be written in component form according to:

$$\begin{aligned}
m\left[\dot{u} - vr + wq - x_g(q^2+r^2) + y_g(pq - \dot{r}) + z_g(pr + \dot{q})\right] &= X_H + X \\
m\left[\dot{v} - wp + ur - y_g(r^2+p^2) + z_g(qr - \dot{p}) + x_g(qp + \dot{r})\right] &= Y_H + Y \\
m\left[\dot{w} - uq + vp - z_g(p^2+q^2) + x_g(rp - \dot{q}) + y_g(rq + \dot{p})\right] &= Z_H + Z \\
I_x\dot{p} + (I_z - I_y)qr - (\dot{r} + pq)I_{xz} + (r^2 - q^2)I_{yz} + (pr - \dot{q})I_{xy} & \\
+m\left[y_g(\dot{w} - uq + vp) - z_g(\dot{v} - wp + ur)\right] &= K_H + K \\
I_y\dot{q} + (I_x - I_z)rp - (\dot{p} + qr)I_{xy} + (p^2 - r^2)I_{zx} + (qp - \dot{r})I_{yz} & \\
+m\left[z_g(\dot{u} - vr + wq) - x_g(\dot{w} - uq + vp)\right] &= M_H + M \\
I_z\dot{r} + (I_y - I_x)pq - (\dot{q} + rp)I_{yz} + (q^2 - p^2)I_{xy} + (rq - \dot{p})I_{zx} & \\
+m\left[x_g(\dot{v} - wp + ur) - y_g(\dot{u} - vr + wq)\right] &= N_H + N
\end{aligned} \tag{2.21}$$

For a deeply submerged vehicle the hydrodynamic forces and moments will be due to added mass and damping, while the hydrostatic forces and moments are due to weight and buoyancy. This suggests that:

$$\underbrace{\mathbf{M}_{RB}\dot{\nu} + \mathbf{C}_{RB}(\nu)\nu}_{\text{rigid-body terms}} + \underbrace{\mathbf{M}_A\dot{\nu} + \mathbf{C}_A(\nu)\nu + \mathbf{D}(\nu)\nu}_{\text{hydrodynamic terms}} + \underbrace{\mathbf{g}(\eta)}_{\text{hydrostatic terms}} = \tau \tag{2.22}$$

such that:

$$\mathbf{M}\dot{\nu} + \mathbf{C}(\nu)\nu + \mathbf{D}(\nu)\nu + \mathbf{g}(\eta) = \tau \tag{2.23}$$

where $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$, $\mathbf{C}(\nu) = \mathbf{C}_{RB}(\nu) + \mathbf{C}_A(\nu)$, and:

$\mathbf{M}$	- system inertia matrix (including added mass)
$\mathbf{C}(\nu)$	- Coriolis-centripetal matrix (including added mass)
$\mathbf{D}(\nu)$	- damping matrix
$\mathbf{g}(\eta)$	- vector of gravitational/buoyancy forces and moments

For UUVs the constant zero frequency added mass terms are used. The matrices are:

$$\mathbf{M}_A = \mathbf{M}_A^\top = -\begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix} \tag{2.24}$$

$$\mathbf{C}_A(\nu) = -\mathbf{C}_A(\nu)^\top = \begin{bmatrix} 0 & 0 & 0 & 0 & -a_3 & a_2 \\ 0 & 0 & 0 & a_3 & 0 & -a_1 \\ 0 & 0 & 0 & -a_2 & a_1 & 0 \\ 0 & -a_3 & a_2 & 0 & -b_3 & b_2 \\ a_3 & 0 & -a_1 & b_3 & 0 & -b_1 \\ -a_2 & a_1 & 0 & -b_2 & b_1 & 0 \end{bmatrix} \tag{2.25}$$

where

$$\begin{aligned} a_1 &= X_{\dot{u}}u + X_{\dot{v}}v + X_{\dot{w}}w + X_{\dot{p}}p + X_{\dot{q}}q + X_{\dot{r}}r \\ a_2 &= Y_{\dot{u}}u + Y_{\dot{v}}v + Y_{\dot{w}}w + Y_{\dot{p}}p + Y_{\dot{q}}q + Y_{\dot{r}}r \\ a_3 &= Z_{\dot{u}}u + Z_{\dot{v}}v + Z_{\dot{w}}w + Z_{\dot{p}}p + Z_{\dot{q}}q + Z_{\dot{r}}r \\ b_1 &= K_{\dot{u}}u + K_{\dot{v}}v + K_{\dot{w}}w + K_{\dot{p}}p + K_{\dot{q}}q + K_{\dot{r}}r \\ b_2 &= M_{\dot{u}}u + M_{\dot{v}}v + M_{\dot{w}}w + M_{\dot{p}}p + M_{\dot{q}}q + M_{\dot{r}}r \\ b_3 &= N_{\dot{u}}u + N_{\dot{v}}v + N_{\dot{w}}w + N_{\dot{p}}p + N_{\dot{q}}q + N_{\dot{r}}r \end{aligned} \tag{2.26}$$

The gravitational force will act through the CG while the buoyancy force will act through the center of buoyancy (CB) defined by $\mathbf{r}_b^b = [x_b, y_b, z_b]^\top$. For an UUV, the submerged weight of the body W and buoyancy force B are given by:

$$W = mg, \quad B = \rho g \nabla \tag{2.27}$$

where ∇ is the volume of fluid displaced by the vehicle, g is the acceleration of gravity (positive downwards), and ρ is the water density. It can be shown that [2]:

$$\mathbf{g}(\eta) = \begin{bmatrix} & (W-B)\ \sin\theta & & \\ - & (W-B)\ \cos\theta\sin\phi & & \\ - & (W-B)\ \cos\theta\cos\phi & & \\ - & (y_gW - y_bB)\ \cos\theta\cos\phi & + & (z_gW - z_bB)\ \cos\theta\sin\phi \\ & (z_gW - z_bB)\ \sin\theta & + & (x_gW - x_bB)\ \cos\theta\cos\phi \\ - & (x_gW - x_bB)\ \cos\theta\sin\phi & - & (y_gW - y_bB)\ \sin\theta \end{bmatrix} \tag{2.28}$$

The damping forces for UUVs can be written as the sum of a linear damping term $\mathbf{D}$ and a nonlinear damping term $\mathbf{D}_n(\nu)$ such that:

$$\mathbf{D}(\nu) = \mathbf{D} + \mathbf{D}_n(\nu) \tag{2.29}$$

where

$$\mathbf{D} = \mathbf{D}^\top = -\begin{bmatrix} X_u & X_v & X_w & X_p & X_q & X_r \\ Y_u & Y_v & Y_w & Y_p & Y_q & Y_r \\ Z_u & Z_v & Z_w & Z_p & Z_q & Z_r \\ K_u & K_v & K_w & K_p & K_q & K_r \\ M_u & M_v & M_w & M_p & M_q & M_r \\ N_u & N_v & N_w & N_p & N_q & N_r \end{bmatrix} \tag{2.30}$$

The nonlinear manoeuvring coefficients $\mathbf{D}_n(\nu)$ are usually modelled by using a 3rd-order Taylor series expansion or modulus functions (quadratic drag). If the xz-plane is a plane of symmetry (starboard/port symmetry) an odd Taylor series expansion containing 1st and 3rd order terms in velocity will be sufficient to describe most manoeuvres.

2.2.4 Equations of Motion Including Ocean Currents

When simulating UUVs it is necessary to include environmental disturbances. For deeply submerged vessels wave-induced disturbances can usually be neglected. Hence, the only environmental load will be ocean currents. Ocean currents are horizontal and vertical circulating systems of ocean waters produced by gravity, wind friction, and water density variation in different parts of the ocean.

The effect of ocean currents can be incorporated in the 6 DOF equations of motion (2.23) by using the concept of relative motion under the assumption of an irrotational flow. Let:

$$\nu_r = [u - u_c^b, v - v_c^b, w - w_c^b, p, q, r]^\top \tag{2.31}$$

where u_c^b, v_c^b, and w_c^b are the body-fixed current velocities. The rigid-body dynamics and generalised hydrodynamic forces are written:

$$\underbrace{\mathbf{M}_{RB}\dot{\nu} + \mathbf{C}_{RB}(\nu)\nu}_{\text{rigid-body terms}} + \underbrace{\mathbf{M}_A\dot{\nu}_r + \mathbf{C}_A(\nu_r)\nu_r + \mathbf{D}(\nu_r)\nu_r}_{\text{hydrodynamic terms}} + \underbrace{\mathbf{g}(\eta)}_{\text{hydrostatic terms}} = \tau \tag{2.32}$$

It is common to assume that the current velocity vector is slowly-varying, that is $\dot{\nu}_c \approx \mathbf{0}$ such that $\dot{\nu}_r \approx \mathbf{0}$. Hence, the equations of motion with currents becomes:

$$\mathbf{M}\dot{\nu} + \mathbf{C}_{RB}(\nu)\nu + \mathbf{C}_A(\nu_r)\nu_r + \mathbf{D}(\nu_r)\nu_r + \mathbf{g}(\eta) = \tau \tag{2.33}$$

The current speed V_c is usually defined in the Earth-fixed reference frame using flow axes, that is $[V_c, 0, 0]^\top$ such that the current speed is directed in the x-direction. The transformation from flow axes to NED velocities can be performed by defining α_c as the flow angle of attack and β_c as the flow sideslip angle. Hence, the 3D current velocities are found by performing two principal rotations [2]:

$$\begin{bmatrix} u_c^n \\ v_c^n \\ w_c^n \end{bmatrix} = \mathbf{R}_{y,\alpha_c}^\top \mathbf{R}_{z,-\beta_c}^\top \begin{bmatrix} V_c \\ 0 \\ 0 \end{bmatrix} \tag{2.34}$$

where the rotation matrices $\mathbf{R}_{y,\alpha_c}$ and $\mathbf{R}_{z,-\beta_c}$ are defined as:

$$\mathbf{R}_{y,\alpha_c} = \begin{bmatrix} \cos\alpha_c & 0 & \sin\alpha_c \\ 0 & 1 & 0 \\ -\sin\alpha_c & 0 & \cos\alpha_c \end{bmatrix}, \quad \mathbf{R}_{z,-\beta_c} = \mathbf{R}_{z,\beta_c}^\top = \begin{bmatrix} \cos\beta_c & \sin\beta_c & 0 \\ -\sin\beta_c & \cos\beta_c & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2.35}$$

Expanding (2.34) using (2.35), yields:

$$u_c^n = V_c \cos\alpha_c \cos\beta_c \tag{2.36}$$
$$v_c^n = V_c \sin\beta_c \tag{2.37}$$
$$w_c^n = V_c \sin\alpha_c \cos\beta_c \tag{2.38}$$

The NED current velocities are then transformed to body-fixed velocities using the Euler angle rotation matrix. Consequently:

$$\begin{bmatrix} u_c^b \\ v_c^b \\ w_c^b \end{bmatrix} = \mathbf{R}_b^n(\mathbf{\Theta})^\top \begin{bmatrix} u_c^n \\ v_c^n \\ w_c^n \end{bmatrix} \tag{2.39}$$

For the 2D case, the 3D equations (2.36)–(2.38) with $\alpha_c = 0$ reduces to:

$$u_c^n = V_c \cos\beta_c \tag{2.40}$$
$$v_c^n = V_c \sin\beta_c \tag{2.41}$$

since the component w_c^n is not used in the horizontal plane. Hence (2.39) reduces to:

$$u_c^b = V_c \cos(\beta_c - \psi) \tag{2.42}$$
$$v_c^b = V_c \sin(\beta_c - \psi) \tag{2.43}$$

2.2.5 Longitudinal and Lateral Models

For UUVs, the 6 DOF equations of motion can in many cases be divided into two non-interacting (or lightly interacting) subsystems:

- **Longitudinal subsystem:** states (u, w, q) and (n, d, θ)
- **Lateral subsystem:** states (v, p, r) and (e, ϕ, ψ)

This decomposition is good for slender bodies (large length/width ratio) as shown in Figure 2.1. Consequently, it is a common assumption for submarines; see Tinker [6], Gertler and Hagen [7], and Feldman [8]. Other examples are flying vehicles like the NPS AUV II (Healey and Lienard [9]) shown in Figure 2.2 and the MARIUS vehicle (Pascoal et al. [10]).

The longitudinal and lateral decompositions are found by considering a slender body with starboard-port symmetry, implying the system inertia matrix takes the following form:

$$\mathbf{M} = \begin{bmatrix} m_{11} & 0 & m_{13} & 0 & m_{15} & 0 \\ 0 & m_{22} & 0 & m_{24} & 0 & m_{26} \\ m_{31} & 0 & m_{33} & 0 & m_{35} & 0 \\ 0 & m_{42} & 0 & m_{44} & 0 & m_{46} \\ m_{51} & 0 & m_{53} & 0 & m_{55} & 0 \\ 0 & m_{62} & 0 & m_{64} & 0 & m_{66} \end{bmatrix} \tag{2.44}$$

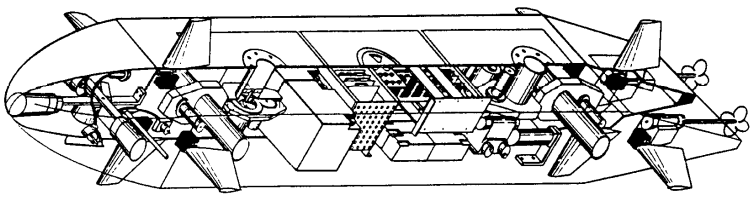

Figure 2.2: The NPS AUV II (Healey and Lienhard 1993).

This gives the two subsystems:

$$\mathbf{M}_{\text{long}} = \begin{bmatrix} m_{11} & m_{13} & m_{15} \\ m_{31} & m_{33} & m_{35} \\ m_{51} & m_{53} & m_{55} \end{bmatrix}, \quad \mathbf{M}_{\text{lat}} = \begin{bmatrix} m_{22} & m_{24} & m_{26} \\ m_{42} & m_{44} & m_{46} \\ m_{62} & m_{64} & m_{66} \end{bmatrix} \tag{2.45}$$

which are evidently decoupled.

Longitudinal Subsystem

Assuming that the states v, p, r, ϕ are small, the longitudinal kinematics for surge, heave, and pitch are, see (2.8) and (2.10):

$$\begin{bmatrix} \dot{d} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} w \\ q \end{bmatrix} + \begin{bmatrix} -\sin\theta \\ 0 \end{bmatrix} u \tag{2.46}$$

For simplicity, it is assumed that higher order damping can be neglected, that is $\mathbf{D}_n(\nu) = \mathbf{0}$. Coriolis is, however, modelled by assuming that $u \gg 0$ and that 2nd-order terms in v, w, p, q, and r are small. Hence, from (2.19) it is seen that:

$$\mathbf{C}_{RB}(\nu)\nu = \begin{bmatrix} m(y_g q + z_g r)p \\ -m(z_g p - v)p \\ m(x_g q - w)u - m(z_g r + x_g p)v \\ -m(x_g q - w)q - m(x_g r + v)r \\ -m(z_g q + u)q + m(x_g p + y_g q)r \\ +m(z_g q + u)w + (I_{yz} q + I_{xz} p - I_z r)p + (-I_{xz} r - I_{xy} q + I_x p)r \end{bmatrix}$$

such that:

$$\mathbf{C}_{RB}(\nu)\nu \approx \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -mu \\ 0 & 0 & mx_g u \end{bmatrix} \begin{bmatrix} u \\ w \\ q \end{bmatrix} \tag{2.47}$$

Notice that $\mathbf{C}_{RB}(\nu) \neq -\mathbf{C}_{RB}^{\top}(\nu)$ for the decoupled model. For simplicity assume that $\mathbf{M}_A$ is diagonal. Hence, the corresponding added mass terms are:

$$\begin{aligned}\mathbf{C}_A(\nu)\nu &= \begin{bmatrix} -Z_{\dot{w}}wq + Y_{\dot{v}}vr \\ -Y_{\dot{v}}vp + X_{\dot{u}}uq \\ (Z_{\dot{w}} - X_{\dot{u}})uw + (N_{\dot{r}} - K_{\dot{p}})pr \end{bmatrix} \\ &\approx \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & X_{\dot{u}}u \\ 0 & (Z_{\dot{w}} - X_{\dot{u}})u & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \end{bmatrix} \end{aligned} \tag{2.48}$$

For a UUV with weight equal to buoyancy ($W = B$) and coincident centres of gravity and buoyancy ($x_g = x_b$), the dynamics then become:

$$\begin{aligned} &\begin{bmatrix} m - X_{\dot{u}} & -X_{\dot{w}} & mz_g - X_{\dot{q}} \\ -X_{\dot{w}} & m - Z_{\dot{w}} & -mx_g - Z_{\dot{q}} \\ mz_g - X_{\dot{q}} & -mx_g - Z_{\dot{q}} & I_y - M_{\dot{q}} \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \end{bmatrix} \\ &+ \begin{bmatrix} -X_u & -X_w & -X_q \\ -Z_u & -Z_w & -Z_q \\ -M_u & -M_w & -M_q \end{bmatrix} \begin{bmatrix} u \\ w \\ q \end{bmatrix} \\ &+ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -(m - X_{\dot{u}})u \\ 0 & (Z_{\dot{w}} - X_{\dot{u}})u & mx_g u \end{bmatrix} \begin{bmatrix} u \\ w \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ W\ BG_z \sin\theta \end{bmatrix} = \begin{bmatrix} \tau_{\mathrm{X}} \\ \tau_{\mathrm{Z}} \\ \tau_{\mathrm{M}} \end{bmatrix} \end{aligned} \tag{2.49}$$

where $BG_z = z_g - z_b$. This model is the basis for forward speed control (state u) and depth/diving autopilot design (states w, q, θ).

If the forward speed is stabilised by a forward speed controller such that:

$$u = u_o = \text{constant} \tag{2.50}$$

forward speed can be eliminated from the longitudinal equations of motion such that:

$$\begin{aligned} &\begin{bmatrix} m - Z_{\dot{w}} & -mx_g - Z_{\dot{q}} \\ -mx_g - Z_{\dot{q}} & I_y - M_{\dot{q}} \end{bmatrix} \begin{bmatrix} \dot{w} \\ \dot{q} \end{bmatrix} + \begin{bmatrix} -Z_w & -Z_q \\ -M_w & -M_q \end{bmatrix} \begin{bmatrix} w \\ q \end{bmatrix} \\ &+ \begin{bmatrix} 0 & -(m - X_{\dot{u}})u_o \\ (Z_{\dot{w}} - X_{\dot{u}})u_o & mx_g u_o \end{bmatrix} \begin{bmatrix} w \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ BG_z W \sin\theta \end{bmatrix} = \begin{bmatrix} \tau_{\mathrm{Z}} \\ \tau_{\mathrm{M}} \end{bmatrix} \end{aligned} \tag{2.51}$$

Moreover, if $\dot{w} = w = 0$ (constant depth) and θ is small such that $\sin\theta \approx \theta$, the *linear pitch dynamics* become:

$$(I_y - M_{\dot{q}})\ddot{\theta} - M_q\dot{\theta} + BG_z W\theta = \tau_{\mathrm{M}} \tag{2.52}$$

where the natural frequency is:

$$\omega_\theta = \sqrt{\frac{BG_z\ W}{I_y - M_{\dot{q}}}} \tag{2.53}$$

Lateral Subsystem

Under the assumption that the states u, w, p, r, ϕ and θ are small, the lateral kinematics, see (2.8) and (2.10), reduce to:

$$\dot{\phi} = p \tag{2.54}$$

$$\dot{\psi} = r \tag{2.55}$$

Again it is assumed that higher order velocity terms can be neglected so that $\mathbf{D}_n(\nu) = \mathbf{0}$, and that the Coriolis terms in $u = u_o$ are the most important. This gives:

$$\mathbf{C}_{RB}(\nu)\nu = \begin{bmatrix} -m(y_g p + w)p + m(z_g r + x_g p)q - m(y_g r - u)r \\ -m(y_g q + z_g r)u + m(y_g p + w)v + m(z_g p - v)w \\ m(x_g r + v)u + m(y_g r - u)v - m(x_g p + y_g q)w \\ \\ +(-I_{yz}q - I_{xz}p + I_z r)q + (I_{yz}r + I_{xy}p - I_y q)r \\ +(-I_{yz}r - I_{xy}p + I_y q)p + (I_{xz}r + I_{xy}q - I_x p)q \end{bmatrix}$$

Hence:

$$\mathbf{C}_{RB}(\nu)\nu \approx \begin{bmatrix} 0 & 0 & mu_o \\ 0 & 0 & 0 \\ 0 & 0 & mx_g u_o \end{bmatrix} \begin{bmatrix} v \\ p \\ r \end{bmatrix} \tag{2.56}$$

Under the assumption of a diagonal $\mathbf{M}_A$, the corresponding added mass terms are:

$$\begin{aligned} \mathbf{C}_A(\nu)\nu &= \begin{bmatrix} Z_{\dot{w}}wp - X_{\dot{u}}ur \\ (Y_{\dot{v}} - Z_{\dot{w}})vw + (M_{\dot{q}} - N_{\dot{r}})qr \\ (X_{\dot{u}} - Y_{\dot{v}})uv + (K_{\dot{p}} - M_{\dot{q}})pq \end{bmatrix} \\ &\approx \begin{bmatrix} 0 & 0 & -X_{\dot{u}}u \\ 0 & 0 & 0 \\ (X_{\dot{u}} - Y_{\dot{v}})u & 0 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \\ r \end{bmatrix} \end{aligned} \tag{2.57}$$

Next, assume that $W = B$, $x_g = x_b$, and $y_g = y_b$. Then the UUV dynamics reduce to:

$$\begin{aligned} &\begin{bmatrix} m - Y_{\dot{v}} & -mz_g - Y_{\dot{p}} & mx_g - Y_{\dot{r}} \\ -mz_g - Y_{\dot{p}} & I_x - K_{\dot{p}} & -I_{zx} - K_{\dot{r}} \\ mx_g - Y_{\dot{r}} & -I_{zx} - K_{\dot{r}} & I_z - N_{\dot{r}} \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \end{bmatrix} \\ &+ \begin{bmatrix} -Y_v & -Y_p & -Y_r \\ -M_v & -M_p & -M_r \\ -N_v & -N_p & -N_r \end{bmatrix} \begin{bmatrix} v \\ p \\ r \end{bmatrix} \\ &+ \begin{bmatrix} 0 & 0 & (m - X_{\dot{u}})u \\ 0 & 0 & 0 \\ (X_{\dot{u}} - Y_{\dot{v}})u & 0 & mx_g u \end{bmatrix} \begin{bmatrix} v \\ p \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ W\ BG_z \sin\phi \\ 0 \end{bmatrix} = \begin{bmatrix} \tau_{\mathrm{Y}} \\ \tau_{\mathrm{K}} \\ \tau_{\mathrm{N}} \end{bmatrix} \end{aligned} \tag{2.58}$$

For vehicles where $\dot{p}$ and p are small (small roll motions) and the speed is $u = u_o$, this reduces

to:

$$\begin{bmatrix} m - Y_{\dot{v}} & mx_g - Y_{\dot{r}} \\ mx_g - Y_{\dot{r}} & I_z - N_{\dot{r}} \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{r} \end{bmatrix} + \begin{bmatrix} -Y_v & -Y_r \\ -N_v & -N_r \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} + \begin{bmatrix} 0 & (m - X_{\dot{u}})u_o \\ (X_{\dot{u}} - Y_{\dot{v}})u_o & mx_g u_o \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} = \begin{bmatrix} \tau_{\mathrm{Y}} \\ \tau_{\mathrm{N}} \end{bmatrix} \tag{2.59}$$

which is the sway-yaw manoeuvring model. The decoupled *linear roll equation* under the assumption of a small ϕ is:

$$(I_x - K_{\dot{p}})\ddot{\phi} - K_p\dot{\phi} + WBG_z\phi = \tau_{\mathrm{K}} \tag{2.60}$$

for which the natural frequency is:

$$\omega_\phi = \sqrt{\frac{BG_z\ W}{I_x - K_{\dot{p}}}} \tag{2.61}$$

2.3 Identification of UUVs

The longitudinal and lateral models presented in Section 2.2.5 involve a large number of parameters. When estimating parameters using system identification (SI) or nonlinear optimisation methods, the main problem is overparametrisation. Moreover, if all the parameters are to be estimated from time-series, it is likely that several of the estimates will dependend on each other due to this overparametrisation. The practical solution to this problem is to exploit a priori information on the nature of the model, and estimate a model of partially known parameters. We will briefly illustrate such an approach in this chapter.

2.3.1 A priori Estimates of Rigid-Body Parameters

For a UUV, the mass parameters are easily obtained since these are computed when constructing the vehicle. Therefore we will assume that the following quantities are known:

$$m, I_x, I_y, I_z, I_{xy}, I_{xz}, I_{yz}, x_g, y_g, z_g \tag{2.62}$$

The moment and product of inertia are usually computed by assuming that the equipment on the vehicle consists of point masses or known geometrical shapes located at known coordinates. A useful tool for this is the parallel axes theorem. Similarly, the displacement and centre of buoyancy must be computed, implying that we know:

$$\nabla, x_b, y_b, z_b \tag{2.63}$$

For a neutrally buoyant vehicle $m = \rho\nabla$.

This means that the elements of:

$$\mathbf{M}_{RB}, \mathbf{C}_{RB}(\nu), \mathbf{g}(\eta) \tag{2.64}$$

are also available a priori.

2.3.2 A priori Estimates of Hydrodynamic Added Mass

For a UUV the hydrodynamic mass terms can be computed by using a CAD/CAM drawing of the vehicle as input to a hydrodynamic software program for instance by using *AutoCad* (www.autocad.com). This data format can usually be transformed to other formats if necessary.

There are several commercially available software programs for computation of hydrodynamic added mass e.g. the 3D-potential theory program of *WAMIT* (www.wamit.com) which assumes that the forward speed is zero (or at least small). This program is quite popular with oil companies since it can be applied to floating structures like rigs and production ships.

For flying vehicles moving at higher forward speeds there are programs that can calculate the elements of $\mathbf{M}_A$ and $\mathbf{C}_A(\nu)$ using 2D-strip theory programs, such as the MARINTEK program *ShipX* (www.marintek.sintef.no) with the add-in for vessel repsonse calculations (*VERES*), which uses the strip theory of Salvesen, Tuck and Faltinsen [11], or SeaWay by Amacom (www.amacom.com).

2.3.3 Identification of Damping Terms

The main problem when designing UUV control systems is to obtain an estimate of the damping terms. One first guess is to use simple formulae for linear damping, and identification methods applied to estimate the remaining terms. For control system design, the formulae for linear damping will be sufficient in many cases.

Formulae for Linear Damping

For a neutrally buoyant vehicle, the formulae for linear damping in *surge, sway, heave* and *yaw* can be derived by assuming that these modes are 2nd-order linear decoupled mass-damper systems:

$$m\ddot{x} + d\dot{x} = \tau \tag{2.65}$$

The linear damping coefficient can be found by specifying the time constant $T > 0$ in:

$$d = \frac{m}{T} \tag{2.66}$$

suggesting that:

$$-X_u \approx \frac{m - X_{\dot{u}}}{T_{surge}} \tag{2.67}$$

$$-Y_v \approx \frac{m - X_{\dot{u}}}{T_{surge}} \tag{2.68}$$

$$-Z_z \approx \frac{m - Z_{\dot{w}}}{T_{heave}} \tag{2.69}$$

$$-N_r \approx \frac{m - N_{\dot{r}}}{T_{yaw}} \tag{2.70}$$

The time constants can be found by performing step responses in surge, sway, heave, and yaw with the UUV.

The remaining modes, *roll* and *pitch*, are assumed to be 2nd-order mass-damper-spring systems:

$$m\ddot{x} + d\dot{x} + kx = \tau \tag{2.71}$$

where

$$\begin{aligned}\frac{d}{m} &= 2\zeta\omega_n \\ &= 2\zeta\sqrt{\frac{k}{m}}\end{aligned} \tag{2.72}$$

or

$$d = 2\zeta\sqrt{km} \tag{2.73}$$

Here, the relative damping ratio ζ can be used to specify the linear damping coefficient d since $\omega_n = \sqrt{k/m}$ is known a priori for known m and k. This gives the following formulae:

$$\begin{aligned}-K_p &\approx 2\zeta_{roll}\ \sqrt{BG_z\ W(I_x - K_{\dot{p}})} \\ -M_q &\approx 2\zeta_{pitch}\ \sqrt{BG_z\ W(I_y - M_{\dot{q}})}\end{aligned}$$

The relative damping ratios ζ_{roll} and ζ_{pitch} for marine vehicles are quite small, typically between 0.1 and 0.2.

Thus an initial estimate of the damping forces and moments will be:

$$\mathbf{D} = -\mathrm{diag}\{X_u, Y_v, Z_w, K_p, M_q, N_r\} \tag{2.74}$$

Identification of Damping Terms using a Partially Known Model

The unknown damping terms, both linear and nonlinear, can be estimated using SI or nonlinear optimisation methods. We will illustrate this by computing both sets of damping terms in Matlab.

INFANTE AUV Model The case study which follows is carried out on a simulation model of the Infante AUV, depicted in Figure 2.3. This craft was designed, developed and modelled at the Instituto Superior Tecnico in Lisbon, Portugal. It is a flying vehicle, making it ideal for the identification of quadratic damping here. For thorough information on the vehicle's development, testing and verification, see Silvestre and Pascoal [12], [13].

Consider the sway-yaw system with nonlinear quadratic damping:

$$\begin{aligned}&\begin{bmatrix} m - Y_{\dot{v}} & mx_g - Y_{\dot{r}} \\ mx_g - Y_{\dot{r}} & I_z - N_{\dot{r}} \end{bmatrix}\begin{bmatrix}\dot{v} \\ \dot{r}\end{bmatrix} + \begin{bmatrix} -Y_v & -Y_r \\ -N_v & -N_r \end{bmatrix}\begin{bmatrix} v \\ r \end{bmatrix} \\ &\quad + \begin{bmatrix} -Y_{|v|v}\,|\nu| & -Y_{|r|r}\,|\nu| \\ -N_{|v|v}\,|\nu| & -N_{|r|r}\,|\nu| \end{bmatrix}\begin{bmatrix} v \\ r \end{bmatrix} + \begin{bmatrix} 0 & (m - X_{\dot{u}})u_0 \\ (X_{\dot{u}} - Y_{\dot{v}})u_0 & (mx_g - Y_{\dot{r}})u_0 \end{bmatrix}\begin{bmatrix} v \\ r \end{bmatrix} = \begin{bmatrix}\tau_Y \\ \tau_N\end{bmatrix}\end{aligned} \tag{2.75}$$

Assume that the only unknowns are the 8 damping terms:

$$-Y_v,\ -Y_r,\ -N_v,\ -N_r,\ -Y_{|v|v},\ -Y_{|r|r},\ -N_{|v|v},\ -N_{|r|r}$$

Least Squares Optimisation It is elementary to parameterise the system according to

$$\mathbf{z} = \phi^\top \theta^* \tag{2.76}$$

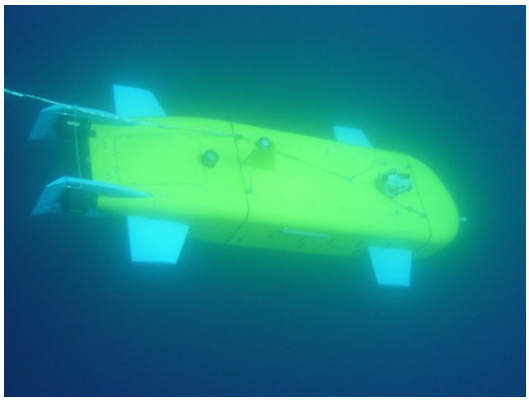

Figure 2.3: The Infante AUV (Picture courtesy of Silvestre and Pascoal [30], [31])

where the terms are defined as follows:

$$\mathbf{y} = [\mathbf{M}\dot{\nu} + \mathbf{C}(\nu)\nu - \tau] \in \mathbb{R}^2 \tag{2.77}$$

$$\mathbf{z} = \frac{\mathbf{y}}{\lambda(s)} \tag{2.78}$$

$$\phi = \begin{bmatrix} -\frac{\nu^\top}{\lambda(s)} & 0 & -\frac{|\nu|\nu^\top}{\lambda(s)} & 0 \\ 0 & -\frac{\nu^\top}{\lambda(s)} & 0 & -\frac{|\nu|\nu^\top}{\lambda(s)} \end{bmatrix}^\top \in \mathbb{R}^{8\times 2} \tag{2.79}$$

$$\theta^* = \begin{bmatrix} Y_v & Y_r & N_v & N_r & Y_{|v|v} & Y_{|r|r} & N_{|v|v} & N_{|r|r} \end{bmatrix}^\top \in \mathbb{R}^8 \tag{2.80}$$

Essentially the system is now in a linear parameterised form, with all the unknowns contained within the parameter vector θ^*. A stable filter $\lambda(s)$ is used to ensure that all signals are available, and therefore that the algorithm is implementable. There are many methods available, both online and offline, to find the best solution to this problem. To take an example of offline estimation define the cost function to be minimised as

$$\min_\theta J = \frac{1}{2}\int_0^T \left(\mathbf{z} - \phi^\top\theta\right)^\top \left(\mathbf{z} - \phi^\top\theta\right) d\tau \tag{2.81}$$

where θ is the calculated estimate of the true parameter vector θ^*. Performing a minimisation with a least squares solution gives convergence to the correct parameters with a residual of the order of $1e^{-3}$. Furthermore, if we take the solution as $T \to \infty$, this result holds in the presence of any zero mean noise. This is an offline methodology: it is rudimentary to take this, and implement a continuous time recursive least squares algorithm, for example the methods described in [14].

Model Based Identification One fairly modern approach to the problem of identification is to use a Lyapunov method, first proposed by Lyshevski and Abel in [15], and further developed in

[16]. This algorithm provides a method of identification for systems of the form:

$$\dot{\mathbf{x}}(t) = \mathbf{A} f[\mathbf{x}(t), \mathbf{u}(t)], \quad t \geq 0, \quad \mathbf{x}(t_0) = \mathbf{x}_0 \tag{2.82}$$

$$\dot{\mathbf{x}}_M(t) = \mathbf{A}_M f[\mathbf{x}_M(t), \mathbf{u}(t)], \quad t \geq 0, \quad \mathbf{x}_M(t_0) = \mathbf{x}_{M0} \tag{2.83}$$

where (2.82) is a parameterisation sectioning the knowns into the function $f(\cdot,\cdot)$ and the unknowns, or only partial knowns, into the matrix A, with A_M as an estimate of this. Equation (2.83) is essentially a mirror of (2.82), except that the state vector is that of the identification model, and not the true states. If we have $x_M = x$ and $A_M = A$, then (2.83) and (2.82) are identical and the problem of estimating the unknowns of A has been solved. The terms are of the following form:

$$\begin{aligned} \mathbf{x},\, \mathbf{x}_M &\in \mathbb{R}^c, \quad u \in \mathbb{R}^m \\ \mathbf{f}(\cdot,\cdot) &\in \mathbb{R}^c \times \mathbb{R}^m \to \mathbb{R}^n \cup C^0 \text{ and is a known analytic vector field} \\ \mathbf{A},\, \mathbf{A}_M &\in \mathbb{R}^{c \times n} \end{aligned}$$

There are three error terms to be considered:

$$\begin{aligned} \Delta\mathbf{x}(t) &= \mathbf{x}(t) - \mathbf{x}_M(t) \\ \Delta\mathbf{A} &= \mathbf{A}(t) - \mathbf{A}_M(t) \\ \Delta\mathbf{f}[\mathbf{x}(t),\, \mathbf{x}_M(t),\, \mathbf{u}(t)] &= \mathbf{f}[\mathbf{x}(t),\, \mathbf{u}(t)] - f[\mathbf{x}_M(t),\, \mathbf{u}(t)] \end{aligned}$$

It is easy to see that $\Delta\mathbf{x}(t)$ is the state error vector: the difference between the actual system states and the states of the identifier, while $\Delta\mathbf{A}(t)$ is the error in the parameter estimation matrix: the difference between the actual parameters and the estimates, and finally $\Delta\mathbf{f}[\mathbf{x}(t),\, \mathbf{x}_M(t),\, \mathbf{u}(t)]$ is the error present in the vector field. Clearly if $\Delta\mathbf{A} = 0$, we have the solution $\mathbf{A} = \mathbf{A}_M$. It has been shown in [16] that a method for evaluating $\mathbf{A}_M$ can be realised using nonlinear error mappings defined such that:

$$\Delta\dot{\mathbf{A}}_M(t) = -\sum_{i=0}^{\eta} \{\Delta\dot{\mathbf{x}}(t) - \mathbf{A}_M(t)\Delta\mathbf{f}[\mathbf{x}(t), \mathbf{x}_M(t), \mathbf{u}(t)]\}^{\frac{2i+1}{2\beta+1}} \mathbf{f}[\mathbf{x}(t), \mathbf{u}(t)]^\top \mathbf{K}(t) \tag{2.84}$$

$$\Delta\mathbf{A}_M(0) = \Delta\mathbf{A}_{M0}$$

where $\mathbf{K}(t) \in \mathbb{R}_+ \to \mathbb{R}^{n \times n}$, $\eta, \beta \in \mathbb{N}$ are chosen to guarantee convergence properties. For the sway/yaw subsystem of (2.75), the parameterisation (2.82) takes the form:

$$\mathbf{A} = \begin{bmatrix} \mathbf{I}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{M}^{-1}\mathbf{N}_L & \mathbf{M}^{-1}\mathbf{N}_Q \end{bmatrix} \tag{2.85}$$

$$\mathbf{f}[\mathbf{x}(t),\, \mathbf{u}(t)] = \begin{bmatrix} (\mathbf{M}^{-1}\tau)^\top & -(\mathbf{M}^{-1}\mathbf{C}\nu)^\top & -\nu^\top & -|\nu|\,\nu^\top \end{bmatrix}^\top \tag{2.86}$$

where N_L is the linear damping matrix from from (2.75) and N_Q is the matrix of quadratic coefficients from the same equation, by taking $|\nu|$ out as a common factor. A postmultiplication of the matrix estimates removes the mass terms from (2.85). After selecting η, $\beta = 2$ and

$$\mathbf{K} = \begin{bmatrix} \mathbf{0}_{4\times4} & \mathbf{0}_{4\times8} \\ \mathbf{0}_{8\times4} & 10^5 \times diag\left\{ \begin{array}{cccccccc} 0.5 & 1 & 0.5 & 1 & 10 & 50 & 10 & 50 \end{array} \right\} \end{bmatrix}$$

simulations were performed for the Infante proceeding forwards at $2\,m/s$ or at just under $4\,knots$,

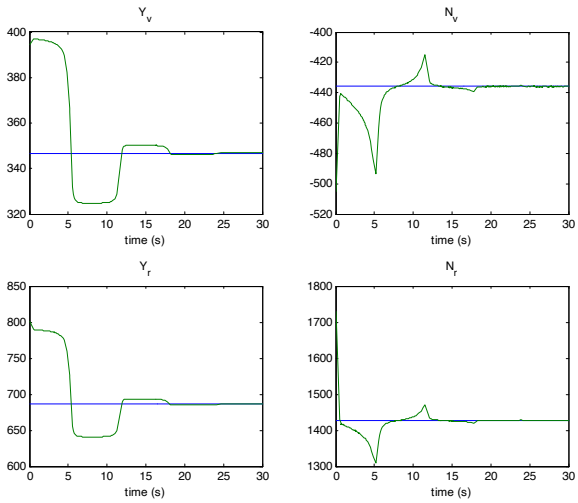

Figure 2.4: Convergence of Linear Damping Estimates

with sinusoidal inputs on both the bow thruster and rudder, and the identification algorithm used to estimate the unknowns in the system. The results of these simulations are presented in Figures (2.4) and (2.5), and show good convergence results to the true parameters.

2.4 Nonlinear Control of UUVs

In this section it is shown how to design autopilots for:

- speed, depth, and pitch control
- heading control

using the longitudinal and lateral equations of motion.

When deriving the multivariable nonlinear backstepping controller the following theorem will be used:

Theorem 1 (MIMO Nonlinear Backstepping Controller) *Consider the nonlinear plant:*

$$\dot{\eta} = \mathbf{J}(\eta)\nu \tag{2.87}$$

$$\mathbf{M}\dot{\nu} + \mathbf{C}(\nu)\nu + \mathbf{D}(\nu)\nu + \mathbf{g}(\eta) = \tau \tag{2.88}$$

$$\tau = \mathbf{B}\mathbf{u} \tag{2.89}$$

and assume that $\mathbf{B}\mathbf{B}^\top > 0$. *Then the nonlinear backstepping controller:*

$$\tau = \mathbf{M}\dot{\nu}_r + \mathbf{C}(\nu)\nu_r + \mathbf{D}(\nu)\nu_r + \mathbf{g}(\eta) - \mathbf{J}^\top(\eta)\mathbf{K}_p\tilde{\eta} - \mathbf{J}^\top(\eta)\mathbf{K}_d\mathbf{s} \tag{2.90}$$

$$\mathbf{u} = \mathbf{B}^\dagger\tau \tag{2.91}$$

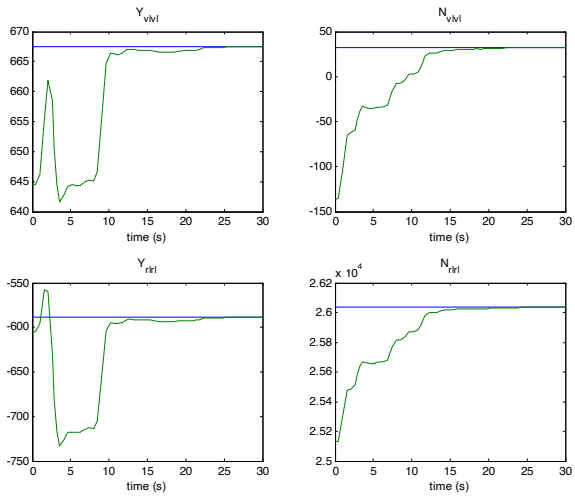

Figure 2.5: Convergence of Quadratic Damping Estimates

with $\mathbf{K}_p = \mathbf{K}_p^\top > 0, \mathbf{K}_d > 0$, $\tilde{\eta} = \eta - \eta_d$ *is the tracking error, and:*

$$\mathbf{s} = \dot{\eta} - \dot{\eta}_r = \dot{\tilde{\eta}} + \mathbf{\Lambda}\tilde{\eta} \tag{2.92}$$

and the reference trajectory (η_d, ν_d) *satisfies* $\dot{\eta}_d = \mathbf{J}(\eta)\nu_d$ *such that:*

$$\dot{\eta}_r = \dot{\eta}_d - \mathbf{\Lambda}\tilde{\eta}, \qquad \mathbf{\Lambda} > 0 \tag{2.93}$$

$$\nu_r = \mathbf{J}^{-1}(\eta)\dot{\eta}_r, \qquad \theta \neq \pm 90^o \tag{2.94}$$

results in an exponentially stable equilibrium point $(\tilde{\eta}, \mathbf{s}) = (\mathbf{0}, \mathbf{0})$. *Consequently,* $\tilde{\eta} \to \mathbf{0}$ *and* $\tilde{\nu} \to \mathbf{0}$ *as* $t \to \infty$.

Proof. The proof is found in Chapter 7 in Fossen [2] where the control Lyapunov function (CLF):

$$V_1 = \frac{1}{2}\tilde{\eta}^\top \mathbf{K}_p \tilde{\eta}, \quad \mathbf{K}_p = \mathbf{K}_p^\top > 0 \tag{2.95}$$

$$V_2 = \frac{1}{2}\mathbf{s}^\top \mathbf{M}^*(\eta)\mathbf{s} + V_1, \quad \mathbf{M}^* = (\mathbf{M}^*)^\top > 0 \tag{2.96}$$

with $\mathbf{M}^*(\eta) = \mathbf{J}^{-\top}(\eta)\mathbf{M}\mathbf{J}^{-1}(\eta)$ is applied. It is straightforward to show that vectorial backstepping in 2 steps gives:

$$\dot{V}_2 = -\mathbf{s}^\top(\mathbf{D}^*(\nu, \eta) + \mathbf{K}_d)\mathbf{s} - \tilde{\eta}^\top \mathbf{K}_p \mathbf{\Lambda}\tilde{\eta}$$

where $\mathbf{D}^*(\nu, \eta) = \mathbf{J}^{-\top}(\eta)\mathbf{D}(\nu)\mathbf{J}^{-1}(\eta)$. This proves that that the equilibrium point $(\tilde{\eta}, \mathbf{s}) = (\mathbf{0}, \mathbf{0})$ is exponentially stable. ■

2.4.1 Speed, Depth and Pitch Control

For UUVs the longitudinal model (2.49) can be used to control speed, depth, and pitch using the three controls τ_1, τ_3 and τ_5, or just speed and depth using τ_1 and τ_5 as control input. In this section, we will discuss both approaches using feedback linearisation and backstepping (Krstic et al. [17]):

- Simultaneous speed, depth and pitch control using three controls.
- Speed and depth control using two controls (underactuated UUV)

The solution for the underactuated UUV will be based on the pure pitching equation (2.52) instead of (2.49). Speed will be controlled independently of these modes. This is based on the assumption that the UUV is stable in heave. The analysis for the underactuated UUV can be extended to included both the surge, heave, and pitch dynamics by following the approach of Pettersen et. al. in [18].

Speed, Depth and Pitch Control using Three Controls

Consider the longitudinal UUV dynamics (2.52) which can be written:

$$\dot{\eta} = \mathbf{J}(\eta)\nu \tag{2.97}$$

$$\mathbf{M}\dot{\nu} + \mathbf{N}(\nu)\nu + \mathbf{g}(\eta) = \tau \tag{2.98}$$

where the coriolis and drag terms are both contained within $\mathbf{N}(\nu)$ and the matrices and vectors are defined as follows:

$$\mathbf{J}(\eta) = \begin{bmatrix} \mathrm{c}\psi\mathrm{c}\theta & \mathrm{c}\psi\mathrm{s}\theta & 0 \\ -\mathrm{s}\theta & \mathrm{c}\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2.99}$$

$$\mathbf{M} = \begin{bmatrix} m - X_{\dot{u}} & -X_{\dot{w}} & mz_g - X_{\dot{q}} \\ -X_{\dot{w}} & m - Z_{\dot{w}} & -mx_g - Z_{\dot{q}} \\ mz_g - X_{\dot{q}} & -mx_g - Z_{\dot{q}} & I_y - M_{\dot{q}} \end{bmatrix} \tag{2.100}$$

$$\mathbf{N}(\nu) = \begin{bmatrix} -X_u & -X_w & -X_q \\ -Z_u & -Z_w & -(m - X_{\dot{u}})u - Z_q \\ -M_u & (Z_{\dot{w}} - X_{\dot{u}})u - M_w & mx_g u - M_q \end{bmatrix} \tag{2.101}$$

$$\mathbf{g}(\eta) = \begin{bmatrix} 0 \\ 0 \\ W\ BG_z \sin\theta \end{bmatrix} \tag{2.102}$$

When applying Theorem 1 we will not implement position feedback in surge since only forward speed is of interest here. Hence the controller must be implemented using velocity control in surge and position control in heave and pitch. The speed command is u_{ref} while the depth and pitch commands are denoted d_{ref} and θ_{ref}. These commands are fed into a reference model exploiting the property $\dot{\eta}_d = \mathbf{J}(\eta)\nu_d$ where we delete the surge position (only velocity feedback in surge). Consequently,

$$\begin{bmatrix} \dot{d}_d \\ \dot{\theta}_d \end{bmatrix} = \begin{bmatrix} -\mathrm{s}\theta & \mathrm{c}\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_d \\ w_d \\ q_d \end{bmatrix} \tag{2.103}$$

where the desired velocities $\nu_d = [u_d, w_d, q_d]^\top$ are computed using the linear filters:

$$\dot{u}_d + \frac{1}{T_u} u_d = \frac{1}{T_u} u_{\text{ref}} \tag{2.104}$$

$$\dot{w}_d + 2\zeta_w \omega_w w_d + \omega_w^2 d_d = \omega_w^2 d_{\text{ref}} \tag{2.105}$$

$$\dot{q}_d + 2\zeta_q \omega_q q_d + \omega_q^2 \theta_d = \omega_q^2 \theta_{\text{ref}} \tag{2.106}$$

where $T_u, \zeta_w, \omega_w, \zeta_q, \omega_q$ are design parameters used to obtain the desired UUV response. Notice that the reference model must preserve the property $\dot{\eta}_d = \mathbf{J}(\eta)\nu_d$ in order to obtain exponential stability.

The controller is chosen according to Theorem 1 with $\mathbf{K}_d = \mathbf{J}^{-\top}(\eta)\mathbf{K}_d^* \mathbf{J}^{-1}(\eta)$ such that:

$$\begin{aligned} \tau &= \mathbf{M}\dot{\nu}_r + \mathbf{N}(\nu)\nu_r + \mathbf{g}(\eta) - \mathbf{J}^\top(\eta)\mathbf{K}_p\tilde{\eta} - \mathbf{K}_d^*\mathbf{J}^{-1}(\eta)\mathbf{s} \\ \mathbf{u} &= \mathbf{B}^\dagger \tau \end{aligned} \tag{2.107}$$

If the controls are a propeller producing the thrust T, and two stern rudders with deflections δ_{S1} and δ_{S2} respectively, the input model becomes:

$$\begin{bmatrix} \tau_1 \\ \tau_3 \\ \tau_5 \end{bmatrix} = \underbrace{\begin{bmatrix} b_{11} & 0 & 0 \\ 0 & b_{22} & b_{23} \\ 0 & b_{32} & b_{33} \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} T \\ \delta_{S1} \\ \delta_{S2} \end{bmatrix} \tag{2.108}$$

which is invertible for physical values of the parameters b_{ij}, that is $\mathbf{B}^\dagger = \mathbf{B}^{-1}$ exists.

The controller gains are modified to obtain speed control in surge. Moreover:

$$\mathbf{K}_p = \text{diag}\{0, K_p^{heave}, K_p^{pitch}\} \tag{2.109}$$

$$\mathbf{K}_d^* = \text{diag}\{K_d^{surge}, K_d^{heave}, K_d^{pitch}\} \tag{2.110}$$

$$\mathbf{\Lambda} = \text{diag}\{0, \lambda^{heave}, \lambda^{pitch}\} \tag{2.111}$$

such that:

$$\begin{aligned} \nu_r &= \mathbf{J}^{-1}(\eta)\dot{\eta}_r \\ &= \mathbf{J}^{-1}(\eta)(\dot{\eta}_d - \mathbf{\Lambda}\tilde{\eta}) \\ &= \nu_d - \mathbf{J}^{-1}(\eta)\mathbf{\Lambda}\tilde{\eta} \end{aligned} \tag{2.112}$$

Expanding this expression gives:

$$\begin{bmatrix} u_r \\ w_r \\ q_r \end{bmatrix} = \begin{bmatrix} u_d \\ w_d \\ q_d \end{bmatrix} + \begin{bmatrix} \mathrm{c}\psi\mathrm{c}\theta & \mathrm{c}\psi\mathrm{s}\theta & 0 \\ -\mathrm{s}\theta & \mathrm{c}\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \lambda^{heave}(d - d_d) \\ \lambda^{pitch}(\theta - \theta_d) \end{bmatrix} \tag{2.113}$$

The feedback term $\mathbf{K}_d^*\mathbf{J}^{-1}(\eta)\mathbf{s}$ is rewritten as:

$$\begin{aligned}\mathbf{K}_d^*\mathbf{J}^{-1}(\eta)\mathbf{s} &= \mathbf{K}_d^*\mathbf{J}^{-1}(\eta)(\dot{\eta}-\dot{\eta}_r)\\ &= \mathbf{K}_d^*(\nu-\nu_r)\\ &= \begin{bmatrix} K_d^{surge} & 0 & 0\\ 0 & K_d^{heave} & 0\\ 0 & 0 & K_d^{pitch}\end{bmatrix}\begin{bmatrix} u-u_r\\ w-w_r\\ q-q_r\end{bmatrix}\end{aligned} \tag{2.114}$$

Then it follows from Theorem 1 that $u \to u_d, d \to d_d$, and $\theta \to \theta_d$ exponentially fast.

Decoupled Speed and Depth Control using Two Controls

In many cases only two controls τ_1 and τ_5 are available, that is $\tau_3 = 0$. This problem can be solved by designing two decoupled controllers for speed and depth control at the price of undesirable interactions between the subsystems.

Forward Speed Control: The forward speed controller is usually designed using a decoupled surge model:

$$(m - X_{\dot{u}})\dot{u} + R(u) = T \tag{2.115}$$

where $(m - X_{\dot{u}})$ is the mass of the vehicle included added mass, $R(u)$ is the added resistance, which can be modelled as quadratic drag:

$$R(u) = -X_{|u|u}|u|u \tag{2.116}$$

for a "flying vehicle", and T is control force. Hence, the forward speed controller can be chosen using *feedback linearisation*:

$$T = R(u) - (m - X_{\dot{u}})\left[K_p(u-u_d) + K_i\int_0^t (u-u_d)d\tau\right] \tag{2.117}$$

implying that u converges to $u_d = constant$ for positive controller gains $K_p > 0$ and $K_i > 0$. The resulting error dynamics consist of a 2nd-order linear system:

$$\ddot{z} + K_p\dot{z} + K_i z = 0, \qquad z = \int_0^t (u-u_d)d\tau \tag{2.118}$$

suggesting that $K_p = 2\zeta\omega_o$ and $K_i = \omega_o^2$ where ζ and ω_o can be used as tuning parameters.

Depth Control: The depth controller will be designed under the assumption of perfect speed control, such that:

$$u = u_o = \text{constant} \tag{2.119}$$

Consider (2.46) and (2.52) where the surge velocity u is treated as a known signal:

$$\begin{bmatrix}\dot{d}\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}\cos\theta & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix} w\\ q\end{bmatrix} + \begin{bmatrix}-\sin\theta\\ 0\end{bmatrix}u \tag{2.120}$$

$$\begin{bmatrix} m - Z_{\dot{w}} & -mx_g - Z_{\dot{q}} \\ -mx_g - Z_{\dot{q}} & I_y - M_{\dot{q}} \end{bmatrix} \begin{bmatrix} \dot{w} \\ \dot{q} \end{bmatrix} + \begin{bmatrix} -Z_w & -Z_q \\ -M_w & -M_q \end{bmatrix} \begin{bmatrix} w \\ q \end{bmatrix}$$
$$+ \begin{bmatrix} 0 & -(m - X_{\dot{u}})u_o \\ (Z_{\dot{w}} - X_{\dot{u}})u_o & mx_g u_o \end{bmatrix} \begin{bmatrix} w \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ BG_z W \sin\theta \end{bmatrix} = \begin{bmatrix} \tau_{\mathrm{Z}} \\ \tau_{\mathrm{M}} \end{bmatrix}$$

If a single stern rudder is used for depth control, the input model becomes:

$$\begin{bmatrix} \tau_{\mathrm{Z}} \\ \tau_{\mathrm{M}} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \delta_S \tag{2.121}$$

where δ_S is the stern rudder angle and b_1 and b_2 are two parameters. This problem can be solved by using the sliding mode controller of Healey and Lienard [9] or backstepping design for underactuated vehicles (Fossen et al. [19]), for instance.

An even simpler approach is to replace the coupled heave and pitch equation with the pure pitching motion such that the governing model becomes:

$$\dot{d} = w\cos\theta - u\sin\theta, \qquad \theta \neq \pm 90^o \tag{2.122}$$
$$\dot{\theta} = q \tag{2.123}$$
$$(I_y - M_{\dot{q}})\dot{q} - M_q q + BG_z W \sin\theta = b_2 \delta_S \tag{2.124}$$

This is of course based on the assumption that the heave dynamics are stable even for the inputs $\tau_3 = b_1\delta_S$ such that we can use $\tau_5 = b_2\delta_S$ for depth control. Backstepping will now be applied to produce the control law for δ_S.

Step 1:
Let d_d be the desired depth. The overall control objective is $d = d_d$ but this is not straightforward, since backstepping with $\dot{d} = w\cos\theta - u\sin\theta$ suggests that $u\sin\theta$ should be used as virtual control in a 3 step procedure. This problem can be circumvented by redefining the output to be controlled as a *sliding surface*:

$$z_1 = (d - d_d) + \lambda\theta, \qquad \lambda > 0 \tag{2.125}$$

It is seen that $z_1 = 0$ only if $\theta = 0$ and $d = d_d$ since $\theta \neq 0$ changes the depth and consequently d moves away from $d_{d.}$.

Consider the control Lyapunov function (CLF):

$$V_1 = \frac{1}{2} z_1^2 \tag{2.126}$$

Consequently,

$$\begin{aligned} \dot{V}_1 &= z_1 \dot{z}_1 \\ &= z_1(w\cos\theta - u\sin\theta - w_d + \lambda q), \qquad w_d = \dot{d}_d \end{aligned} \tag{2.127}$$

Choosing λq as *virtual controller* in Step 1, that is:

$$\lambda q := \alpha_1 + z_2 \tag{2.128}$$

gives:

$$\dot{V}_1 = z_1\left(\alpha_1 + z_2 + w\cos\theta - u\sin\theta - w_d\right) \tag{2.129}$$

The *stabilising function* α_1 is chosen as:

$$\alpha_1 = u\sin\theta - w\cos\theta + w_d - k_1 z_1 \tag{2.130}$$

where $k_1 > 0$ is the feedback gain while the new state z_2 is left undefined until the next step. This finally gives:

$$\dot{V}_1 = -k_1 z_1^2 + z_1 z_2 \tag{2.131}$$

Step 2:
In Step 2, we start with the z_2 dynamics which can be obtained from (2.128). Moreover:

$$\dot{z}_2 = \lambda\dot{q} - \dot{\alpha}_1 \tag{2.132}$$

The next CLF is chosen as.

$$V_2 = V_1 + \frac{1}{2}z_2^2 \tag{2.133}$$

such that:

$$\begin{aligned}\dot{V}_2 &= \dot{V}_1 + z_2\dot{z}_2 \\ &= -k_1 z_1^2 + z_1 z_2 + z_2(\lambda\dot{q} - \dot{\alpha}_1)\end{aligned} \tag{2.134}$$

Substituting the dynamics (2.122) into the expression for $\dot{q}$ gives:

$$\dot{V}_2 = -k_1 z_1^2 + z_2\left(z_1 + \frac{\lambda}{I_y - M_{\dot{q}}}(b_2\delta_S - M_q q + BG_z W\sin\theta) - \dot{\alpha}_1\right) \tag{2.135}$$

Then the control law:

$$\delta_S = \frac{1}{b_2}\left[M_q q - BG_z W\sin\theta - \frac{I_y - M_{\dot{q}}}{\lambda}\left(\dot{\alpha}_1 - z_1 - k_2 z_2^2\right)\right] \tag{2.136}$$

where $k_2 > 0$, finally gives:

$$\dot{V}_2 = -k_1 z_1^2 - k_1 z_2^2 \tag{2.137}$$

Hence, the equilibrium point $(z_1, z_2) = (0, 0)$ is exponentially stable and $d \to d_d$ and $\theta \to 0$. When implementing the control law, the following approximation can be used:

$$\begin{aligned}\dot{\alpha}_1 &= \dot{u}\sin\theta + uq\cos\theta + wq\sin\theta - \dot{w}\cos\theta + \dot{w}_d - k_1\left((\dot{d} - \dot{d}_d) + \lambda\dot{\theta}\right), \\ &\overset{\dot{u}=\dot{w}=0}{\approx} uq\cos\theta + wq\sin\theta + \dot{w}_d - k_1(w\cos\theta - u\sin\theta - w_d + \lambda q)\end{aligned} \tag{2.138}$$

since $\dot{u}$ and $\dot{w}$ are small. Constant depth is obtained by choosing $d_d = constant$ and $\dot{w}_d = w_d = 0$.

2.4.2 Heading Control

The heading control problem is usually solved using a Nomoto approximation (Nomoto et al. [20]) where the roll mode is left uncontrolled. From (2.59), we recognise the manoeuvring model of

Davidson and Schiff [21]:

$$\begin{bmatrix} m - Y_{\dot{v}} & mx_g - Y_{\dot{r}} \\ mx_g - Y_{\dot{r}} & I_z - N_{\dot{r}} \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{r} \end{bmatrix} + \begin{bmatrix} -Y_v & -Y_r \\ -N_v & -N_r \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} + \begin{bmatrix} 0 & (m - X_{\dot{u}})u_o \\ (X_{\dot{u}} - Y_{\dot{v}})u_o & mx_g u_o \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} = \begin{bmatrix} \tau_{\mathrm{Y}} \\ \tau_{\mathrm{N}} \end{bmatrix} \tag{2.139}$$

Assume that the heading is controlled by using a rudder with deflection δ_R such that:

$$\begin{bmatrix} \tau_{\mathrm{Y}} \\ \tau_{\mathrm{N}} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \delta_R \tag{2.140}$$

where b_1 and b_2 are two parameters. Then, the rudder angle can be computed as:

$$\delta_R = \frac{1}{b_2} \tau_{\mathrm{N}} \tag{2.141}$$

where τ_{N} is computed by the control law. The 2nd-order Nomoto models are obtained as:

$$\frac{v}{\delta_R}(s) = \frac{K_v(1 + T_v s)}{(1 + T_1 s)(1 + T_2 s)} \tag{2.142}$$

$$\frac{r}{\delta_R}(s) = \frac{K(1 + T_3 s)}{(1 + T_1 s)(1 + T_2 s)} \tag{2.143}$$

where T_i $(i = 1, \ldots, 3)$ and T_v are time constants, and K and K_v are the gain constants.

A 1st-order approximation to (2.142) is obtained by defining the *effective time constant*:

$$T = T_1 + T_2 - T_3 \tag{2.144}$$

such that:

$$\frac{r}{\delta_R}(s) = \frac{K}{(1 + Ts)} \tag{2.145}$$

where T and K are known as the Nomoto time constant and gain constant, respectively. Including the yaw kinematics:

$$\dot{\psi} = r \tag{2.146}$$

finally yields:

$$\begin{aligned} \frac{\psi}{\delta_R}(s) &= \frac{K(1 + T_3 s)}{s(1 + T_1 s)(1 + T_2 s)} \\ &\approx \frac{K}{s(1 + Ts)} \end{aligned} \tag{2.147}$$

This model is the most popular model for ship autopilot design due to its simplicity and accuracy, and applies well to UUVs as well.

The gain and time constants in Nomoto's 1st- and 2nd-order models can be made invariant with respect to length L and speed $U = \sqrt{u^2 + v^2 + w^2}$ by defining:

$$K' = (L/U)\, K, \qquad T' = (U/L)\, T \tag{2.148}$$

This suggests that the 1st-order ship dynamics can be expressed as:

$$\dot{\psi} = r \tag{2.149}$$

$$\dot{r} = -\frac{1}{T'}\left(\frac{U}{L}\right) r + \frac{K'}{T'}\left(\frac{U}{L}\right)^2 \delta_R \tag{2.150}$$

A robust control law for heading is:

$$\delta_R = \frac{T'}{K'}\left(\frac{L}{U}\right)^2 \left(\tau_{\mathrm{FF}} \underbrace{-K_p e - K_d \dot{e} - K_i \int_0^t e(\tau) d\tau}_{\mathrm{PID}}\right), \qquad e = \psi - \psi_d \tag{2.151}$$

$$\tau_{\mathrm{FF}} = \frac{1}{T'}\left(\frac{U}{L}\right) \dot{r}_d + r_d \tag{2.152}$$

where τ_{FF} is reference feedforward from $r_d = \dot{\psi}_d$.

The controller gains are easily found, for instance by using pole placement. For the case $K_i = 0$ the error dynamics becomes:

$$\ddot{e} + \left(\frac{1}{T'}\left(\frac{U}{L}\right) + K_d\right)\dot{e} + K_p e = 0 \tag{2.153}$$

This suggests that:

$$2\omega_n \zeta = \left(\frac{U}{L}\right)\frac{1}{T'} + K_d \tag{2.154}$$

$$\omega_n^2 = K_p \tag{2.155}$$

where ω_n and ζ are pole placement design parameters. The time constant for the integrator ($K_i \neq 0$) is chosen approximately as $\omega_n/10$. This gives the following pole placement algorithm; see Fossen [2]:

Pole Placement Algorithm for Heading Autopilot

1. Specify the bandwidth $\omega_b > 0$ and the relative damping ratio $\zeta > 0$
2. Compute the natural frequency: $\omega_n = \frac{1}{\sqrt{1-2\zeta^2+\sqrt{4\zeta^4-4\zeta^2+2}}}\omega_b$
3. Compute the P-gain: $K_p = \omega_n^2$
4. Compute the D-gain: $K_d = 2\zeta\omega_n - \left(\frac{U}{L}\right)\frac{1}{T'}$
5. Compute the I-gain: $K_i = \frac{\omega_n}{10} K_p$

2.4.3 Alternative Methods of Control

There are various other methods available for control of UUVs, which cannot, for reasons of space, be presented here. For comparisons of H_∞ control , μ-synthesis, and sliding mode control, one may see Logan's work in [22]. Other comparisons of the various control techniques can be found in [23] and [24]. For additional information on sliding mode control see [25], and for experimental work, [26], which presents the results of using sliding mode control on the MUST UUV. The use of adaptive control can be seen, for instance, in [27], [28] and [29].

2.5 Conclusions

The 6 DOF equations of motion for UUVs have been presented as longitudinal and lateral subsystems for speed, diving, depth, and heading control. Nonlinear optimisation has been used as a tool for determination of hydrodynamic coefficients in conjunction with hydrodynamic software packages. Nonlinear control theory has been applied to combined diving and depth control while heading control is solved using a conventional PID controller with reference feedforward. The presented solutions are intended for UUVs moving at forward speed (flying vehicles).

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